Attention-Enhanced U-Net for Automated Image Colorization

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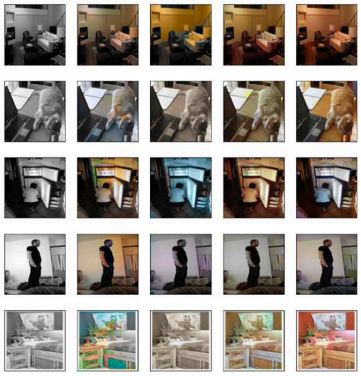

Figure 1. Colorizing monochrome image (leftmost) in comparison with ground truth (rightmost) [4]

1. Task

Our task is to perform fully automated image colorization, transforming grayscale images into visually realistic colored versions. The primary difficulty of this task lies in the multi-modal nature of the problem, where the sky could be of different shades of blue or even gray. Traditional machine learning models often struggle with this ambiguity, frequently producing blurry images by averaging these possibilities.

Our objective is to train a model that outputs vibrant, structurally coherent colorizations. We will evaluate our generated images against ground-truth colored images using quantitative perceptual and reconstruction metrics, as defined in Section 4.

2. Related Work

Image colorization, or the process of adding visually plausible color to a monochrome image, is an area that has been largely automated through algorithms that automatically assign colors based on learned patterns. Recent years have seen the development of a variety of different approaches of different types, including diffusion, transformer, CNN, and GAN-based models.

Foundational CNNs and the LAB Color Space: Early deep learning approaches struggled with the multi-modal nature of colorization, often producing washed-out, sepia-toned images due to the averaging effect of standard Euclidean (L2) loss functions. Zhang et al. (2016) [1] addressed this in their seminal work, Colorful Image Colorization. By converting images to the CIELAB color space and treating color prediction as a classification task over quantized color bins, they successfully encouraged the model to predict vibrant, saturated colors. Their work established the standard practice of using the L channel as the input to predict the a and b channels, a technique we adopt in our pipeline.

Conditional Generative Adversarial Networks (cGANs): The introduction of adversarial training revolutionized image translation. Isola et al. (2017) [2] introduced the Pix2Pix framework in Image-to-Image Translation with Conditional Adversarial Networks. They demonstrated that a cGAN utilizing a U-Net generator and a PatchGAN discriminator could act as a general-purpose solution for translation tasks, including colorization. The adversarial loss effectively penalized visually unrealistic results, while the U-Net's skip connections preserved vital low-level spatial details. This framework forms the structural basis of our proposed generator.

Attention Mechanisms in Image Generation: While standard U-Nets excel at local feature extraction, they often struggle with long-range dependencies (e.g., ensuring a person's left and right sleeves are the same color). The integration of self-attention into generative models, popularized by models like SAGAN (Zhang et al., 2019) [3], allowed networks to compute the response at a given pixel as a weighted sum of the features at all other positions, enforcing global structural and color consistency.

Recent Advances in Attention-Enhanced U-Nets: Recent models have successfully merged these concepts. Nathanael and Prasetyo (2024) [4] demonstrated a Multi-Attention U-Net system that colorized images using attention modules alongside embeddings from a pre-trained PatchGAN classifier. Their architecture yielded performance metrics indicating that multi-attention significantly improved colorization quality compared to standard U-Nets.

However, due to high computational overhead, their model was limited to a short 25-epoch training cycle, limiting its ability to accurately colorize highly complex features like human faces. Our approach aims to balance this complexity by optimizing the Attention U-Net specifically for the LAB color space with targeted loss functions.

3. Approach

To optimize computational efficiency and model stability, we will process images using the CIELAB (LAB) color space rather than the standard RGB format. In this space, the L channel represents lightness (the grayscale input), while the a and b channels represent color values. By isolating the color information, the algorithm only needs to predict two color channels instead of three. These two channels will be scaled to a range of [-1, 1], simplifying the base computation.

Our proposed architecture utilizes a Generative Adversarial Network (GAN) framework, specifically employing an Attention-Enhanced U-Net as the generator. The U-Net structure, which bridges an encoder and decoder, is ideal for preserving spatial information and facilitating pixel-to-pixel learning. We will integrate attention gates into the U-Net's skip connections. This enhancement allows the model to selectively focus on salient, complex features (such as faces or specific objects) while suppressing irrelevant background noise, yielding cleaner and more precise results.

During training, the GAN's discriminator will apply an adversarial loss to encourage vibrant, realistic color distributions. Simultaneously, an L1 reconstruction loss will be applied to the generator to enforce structural accuracy and ensure the output closely preserves the spatial information of the input. The combination of the Attention-Enhanced U-Net and adversarial training is expected to yield sharper, more vivid colorizations than traditional CNNs.

4. Dataset and Metric

We shall use a randomized subset of the ImageNet-1K dataset to train and evaluate our model. To limit the computational intensity in this half-a-semester project, we would restrict our dataset to 10,000 images for training, 1000 for validation and testing respectively.

Images will first be collectively resized to 256 × 256 pixels, then converted from the RGB colorspace to the LAB standard. The L (lightness), a and b (color) channels would be normalized to [-1, 1], with lightness serving as input and the color channels as outputs. Artificial expansion to our dataset to make the model robust against transformations would be done using data augmentation on the training set, such as rotations and horizontal flips.

To verify the effectiveness of our model, training and test images will be collected by pairing colored photographs with an associated grayscale conversion. The primary evaluation metric to be used is the Structural Similarity Index Measure (SSIM), a measure for the perceived similarity between images to the human eye. A secondary metric would be the Peak Signal-to-Noise Ratio (PSNR), a metric used to measure reconstruction quality.

Our goal is to show that our Attention-Enhanced U-Net could achieve higher SSIM and PSNR on the test set as compared to a baseline U-Net model trained solely on the L1 loss.

5. Approximate Timeline

The following is our approximated schedule.

Task	**Deadline**
Implement dataset pipeline with skeleton code	03/16/26
Optimize and train model	04/01/26
[Internal] Finalize model and results	04/14/26
Prepare report and presentation materials	04/14/26
[Internal] Finalize presentation	04/21/26
Presentation Due	04/28/26
Report Due	05/05/26

References

[1] Zhang, R., Isola, P., & Efros, A. A. (2016). Colorful image colorization. In *European conference on computer vision* (pp. 649-666). Springer, Cham.

[2] Isola, P., Zhu, J. Y., Zhou, T., & Efros, A. A. (2017). Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1125-1134).

[3] Zhang, H., Goodfellow, I., Metaxas, D., & Odena, A. (2019). Self-attention generative adversarial networks. In *International Conference on Machine Learning (ICML)* (pp. 7354-7363). PMLR.

[4] Nathaniel, O. T. and Prasetyo, S. Y., Color and Attention for U: Modified Multi Attention U-Net for a Better Image Colorization, *Int. J. Inform. Visualization*, 8(3), 2024.